

	GLOBAL ECONOMICS GRADE: 10 LEARNING EVIDENCE 2 Standard Deviation and Mean (C4) TEACHER'S NAME: Nicolás López Cuéllar	TERM 3 2025–2026
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Learning objective: Interpret expected value, variance, and standard deviation from probability distributions by linking graph shape, center, and spread.	Assessment criteria: C4. Interprets the concept of expected value, variance, and standard deviation of a probability function.
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1 Matching Problems

Problem 1. (1) Uniform: match graphs with their means **Problem 4.** (2) Uniform: match graph, mean, and spread **Problem 9.** (1) Triangular: match graph, mean, and spread

Correct matches:

Graph	Answer
A	−1
B	2
C	5
D	9

Correct matches:

Graph	(μ, σ)
A	$(-1, 1.15)$
B	$(2, 1.73)$
C	$(6, 2.31)$
D	$(9, 1.15)$

Correct matches:

Graph	(μ, σ)
A	$(0, 0.82)$
B	$(4, 1.22)$
C	$(8, 0.41)$

Problem 2. (1) Uniform: match graphs with their spread **Problem 7.** (1) Triangular: same spread, different mean **Problem 10.** (2) Triangular: match graph, mean, and spread

Correct matches:

Graph	Answer
A	0.58
B	1.15
C	1.73
D	2.89

Correct matches:

Graph	Answer
A	$-\frac{4}{3}$
B	$\frac{7}{3}$
C	$\frac{14}{3}$
D	$\frac{28}{3}$

Correct matches:

Graph	(μ, σ)
A	$(-\frac{4}{3}, 0.85)$
B	$(\frac{7}{3}, 1.25)$
C	$(\frac{17}{3}, 1.65)$
D	$(\frac{28}{3}, 0.85)$

Problem 3. (1) Uniform: same mean, different spread match graph, mean, and spread **Problem 8.** (1) Triangular: **Problem 13.** (1) Normal I: match mean and standard deviation

Correct matches:

Graph	(μ, σ)
A	$(2, 1.15)$
B	$(5, 1.73)$
C	$(8, 1.15)$

Correct matches:

Graph	Answer
A	0.41
B	1.08
C	1.47
D	2.04

Correct matches:

Graph	(μ, σ)
A	$(3.2, 0.9)$
B	$(3.9, 1.2)$

Problem 14. (1) Normal II:

Problem 15. (1) Normal III:

match mean and standard de-

match mean and standard de-

viation

viation

Correct matches:

Graph	(μ, σ)
A	$(2.6, 1.1)$
B	$(3.3, 0.9)$
C	$(4.0, 1.3)$

Correct matches:

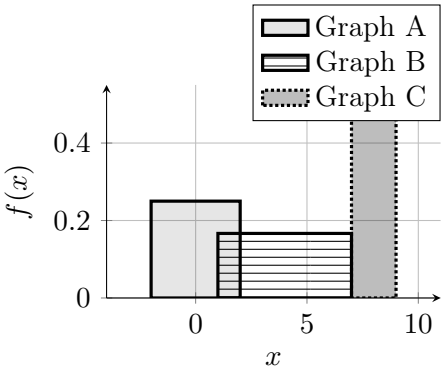
Graph	(μ, σ)
A	$(2.0, 1.4)$
B	$(3.1, 0.9)$
C	$(4.7, 2.0)$
D	$(5.8, 1.1)$

2 Drawing Problems

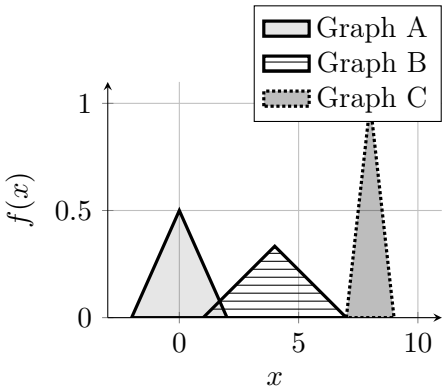
Problem 5. (2) Uniform: draw three uniform distributions

Problem 11. (2) Triangular: draw three triangular distributions

$$A : U[-2, 2], \quad B : U[1, 7], \quad C : U[7, 9].$$



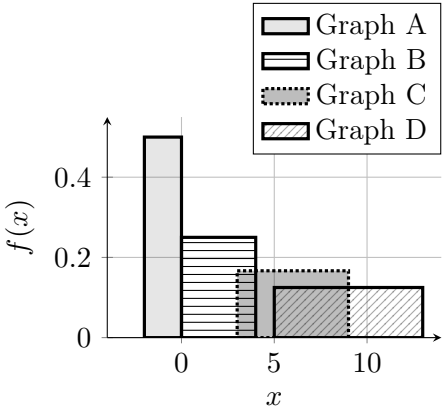
$$A : (a, c, b) = (-2, 0, 2), \quad B : (1, 4, 7), \quad C : (7, 8, 9).$$



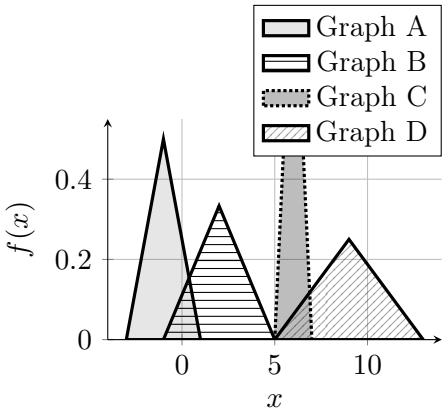
Problem 6. (2) Uniform: draw four uniform distributions

Problem 12. (2) Triangular: draw four triangular distributions

$$A : U[-2, 0], \quad B : U[0, 4], \quad C : U[3, 9], \quad D : U[5, 13].$$

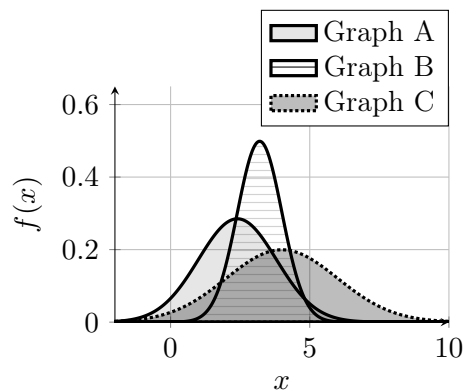
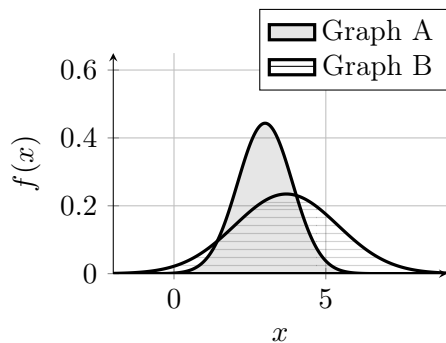


$$A : (a, c, b) = (-3, -1, 1), \quad B : (-1, 2, 5) \\ C : (5, 6, 7), \quad D : (5, 9, 13).$$



Problem 16. (2) Normal IV: draw two normal distributions

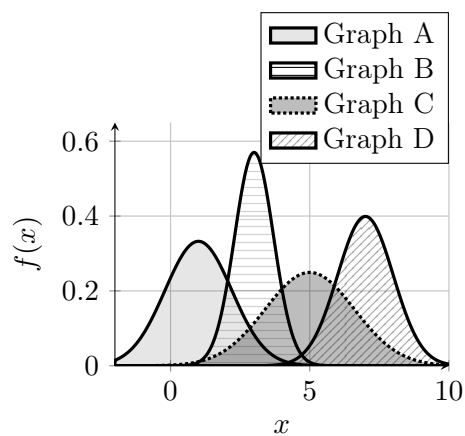
$$A : N(3.0, 0.9^2), \quad B : N(3.7, 1.7^2).$$



Problem 18. (3) Normal VI: draw four normal distributions

$$A : N(1.0, 1.2^2), \quad B : N(3.0, 0.7^2)$$

$$C : N(5.0, 1.6^2), \quad D : N(7.0, 1.0^2).$$



Problem 17. (2) Normal V: draw three normal distributions

$$A : N(2.4, 1.4^2), \quad B : N(3.2, 0.8^2), \quad C : N(4.0, 2.0^2).$$